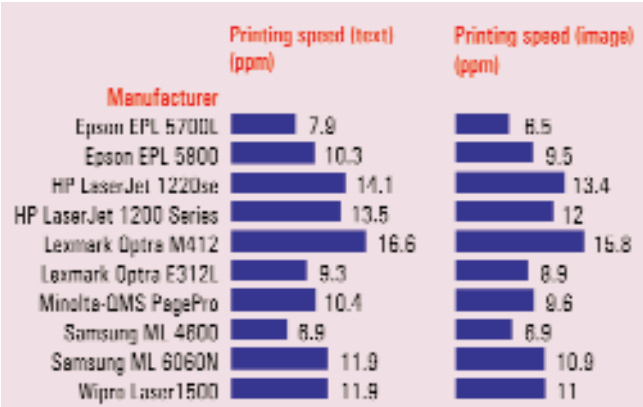




The page per minute (ppm) was arrived at by taking the average of the scores obtained in printing the text and image document

printers were followed by the Wipro Laser 1500, capable of printing 11.01 copies of the image in a minute. The Samsung ML 6060N wasn't able to keep up, printing only 10.85 prints in a minute. The Epson range of printers fared quite badly here. The EPL 5800 was able to print 9.52 copies of the image in a minute, and the Minolta PagePro 1100 gave 9.59 prints a minute.

The printer that stood out, as far as speed goes, was undeniably the **Lexmark Optra M412**, with the HPs following close behind.

Image quality

Graphics image: The Lexmark OptraE312 printer was able to easily replicate details present in the image. The Epson EPL 5700L too was able to bring out similar image quality as the Lexmark, but for the horizontal stripes that were made by the drum of the printer. The printer, when delivered to us for testing, was not in working condition and needed to be serviced by a technician before we actually ran the tests.

The Hewlett-Packard range of printers were able to capture details in all other aspects of the image such as the background, the wrinkles on the face, the folds on the sheet, but when it came to the details of the hair, the printer was not able to replicate the image to its full extent. The Epson EPL 5800 printed the image right but the contrast level in the print was too high as compared to the Epson EPL 5700L.

The Minolta QMS PagePro 1100 printed the image better than the Epson EPL 5800. There was less banding but here too the contrast level in the printout was higher, which was not acceptable. The Wipro

Laser 1500 did just the opposite; here the brightness levels in the prints were higher. This proved bad for the print quality, as the printer was not able to give details in the image.

We also looked out for the ability of the printer to convert the image's colour to greyscale with sufficient variation to represent colours. **The Lexmark Optra E312L** successfully replicated this effect.

This printer would be a right choice for people who need to print pre-views of graphical images or photographs without wasting time and resources.

Vector image: Out of the 10 printers that we received, only five were capable of printing the vector images perfectly.

The Epson EPL 5800 printed the image but the middle part of the image was totally blanked out. So for the quality test, this printer couldn't be included. The printer that excelled in this

particular test was the **Hewlett-Packard LaserJet 1220Se** and **LaserJet 1200**. These printers were able to successfully replicate both the full bleed size image and the thumbnail size image with proper details in both the images.

The Lexmark Optra M412 too printed the vector image but could not properly define the colours. But in terms of the details in the image the printer betters the Epson EPL 5700L.

The Epson EPL couldn't replicate colours, and there was a drop in the quality of the image in the thumbnail-size image. To add to these grievances, there were patches throughout the printout due to scratches on the printer drum. The



EPSON EPL 5800: Not that great for vector printing

Paper Jams

Like inkjet printers, even laser printers are prone to paper jams. This problem usually occurs in heavily used printers. It's mainly due to the rollers in the paper path getting worn out. When new, these rollers that are generally made of rubber contain a certain amount of grip that enable them to hold the paper and pass it forward, but after sometime these rollers wear out, leading to paper jams. Jamming problems may also arise from the dust that collects on the paper path sensors. When a paper passes through the paper path in the printer assembly, the sensors send out signals to other sensors present in the paper path, to tell them the location of the paper. Due to accumulation of dust on these sensors, if even one of the sensors is not able to give proper paper location, the printer might experience paper jams.

Recovering from paper jams

First pull out the paper tray and look into the assembly for jammed paper. If you see jammed paper, gently pull it out and reinsert the paper trays. If you don't see jammed paper, open the back door of the printer. If you see jammed paper around here, gently pull it out and close the door. Else, open the top door and pull out the toner cartridge. Look for paper in the paper path around the fuser. If you see jammed paper, pull it out gently and reinsert the ink cartridge. Extreme caution needs to be exercised—if you try to pull out the paper violently, part of the paper might tear and get stuck in the fuser assembly. If you don't see jammed paper in any of the above mentioned areas, it would make more sense to reinsert the toner cartridge and call a professional.

